

HALO: High Autonomous Low-SWaP Operations

Team Members:

- Sloan Hatter shatter2022@my.fit.edu
- Blake Gisclair bgisclair2022@my.fit.edu

Faculty Advisor: Dr. Ryan T White rwhite@fit.edu

Client: Dr. Ryan T White rwhite@fit.edu

Progress Matrix of Milestone 5

Task	Completion %	Sloan	To Do
Showcase Poster Draft	90%	100%	Add in results and data images
Obtain a CNN model trained on the same dataset as the ViT	100%	100%	None
Compare the performance of the 32-bit CNN model to the performance of the 16-bit quantized ViT model	70%	100%	Gather CNN performance metrics for comparison
Fix 8-bit ViT model	70%	100%	Prime 32-bit ViT model with Q/DQ nodes for 8-bit quantization on the Jetson
Attempt 4-bit quantization	70%	100%	Run specialized commands for 4-bit representation on the Jetson using the primed 32-bit ViT model

Discussion of Accomplished Tasks for Milestone 5:

- **Showcase Poster Draft**
 - X
- **Obtain a CNN model trained on the same dataset as the ViT**
 - X
- **Compare the performance of the 32-bit CNN model to the performance of the 16-bit quantized ViT model**
 - X

- **Fix 8-bit ViT model**
 - X
- **Attempt 4-bit quantization**
 - X

Discussion of Contribution to Milestone 5:

- **Sloan Hatter:**

Task Matrix for Milestone 6:

Task	Sloan
Finalize Showcase Poster	100%
Demo Video	100%
User/Developer Manual	100%
Finalize performance metrics and results for both the quantized 16-bit ViT and the 32-bit CNN	100%
Record metrics for quantized 8-bit ViT for CNN comparison if feasible	100%
Record metrics for quantized 4-bit ViT for CNN comparison if feasible	100%

Discussion of Planned Tasks for Milestone 6:

- **Finalize Showcase Poster**
 - X
- **Demo Video**
 - X
- **User/Developer Manual**
 - X
- **Finalize performance metrics and results for both the quantized 16-bit ViT and the 32-bit CNN**
 - X
- **Record metrics for quantized 8-bit ViT for CNN comparison if feasible**
 - X
- **Record metrics for quantized 4-bit ViT for CNN comparison if feasible**
 - X

Date of Meetings:

- 03/19/26

Client Feedback on Milestone 5:

See Faculty Advisor Feedback below.

Faculty Advisor Feedback on Milestone 5:

Faculty Advisor Signature: _____ Date: 30 March 2026